The Market for Elite Status*

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Abstract

Why does competition not eliminate elite institutions that consume more resources than they return? We price elite status as a club membership in the general equilibrium framework of Ellickson, Grodal, Scotchmer, and Zame. Elite managers earn visible salary premia yet lower payoffs net of elite education: the status discount that finances the system. Elite firms out-produce their rivals firm by firm, yet aggregate output falls as the taste for status spreads, because the elite complex yields less output per unit of resources. Nevertheless, every equilibrium lies in the core: markets work; material efficiency does not follow. If status is valued for its scarcity, an unpriced dilution externality generates over-entry, and a membership tax raises both welfare and output. In an overlapping-generations extension, status is a luxury, and growth finances the status spending that arrests it: societies converge, with full foresight, to steady states permanently poorer than their status-free counterfactuals. **Keywords:** elite status; clubs; general equilibrium; positional externalities; credentialism; elite overproduction.
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1 Introduction

A familiar argument holds that competitive markets discipline waste: an institution that consumes more resources than it returns should be undercut by rivals and disappear. Elite institutions appear to defy this logic. Elite institutions absorb ever more resources per graduate, and how much of what they sell is skill remains contested: for the average admitted student, attending a more selective college adds little to earnings once student quality is held fixed (Dale and Krueger, 2002, with the notable exceptions of disadvantaged

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students, and with returns tracking college resources); the causal payoff of elite programs accrues disproportionately to the already advantaged — in Chile, entirely to men from elite private high schools (Zimmerman, 2019); admission to Ivy-Plus colleges operates through gateways to prestigious firms and graduate schools (Chetty et al., 2026); and among interwar Harvard students the labor-market premium to joining the right social clubs exceeded, descriptively, the premium to academic performance (Michelman et al., 2022). The firms that recruit preferentially from these institutions screen on pedigree and polish as much as on tested ability (Rivera, 2015). Yet elite schools and the firms their graduates run have not been competed away; they have persisted, and expanded, through decades of intense competition in product, labor, and education markets.

This paper's answer is not that competition fails, but that it succeeds. We model elite status as what it observably is: a membership, first in a school and subsequently in the class of people who run certain firms, that is bought and sold at market prices. The natural home for such a model is the general equilibrium theory of clubs developed by Ellickson et al. (1999, 2001, 2006), in which group memberships are priced alongside ordinary commodities and agents care about the characteristics of their fellow members. Within that framework we obtain three results.

First, equilibrium compensation reproduces the configuration compensation data show. Elite managers earn a gross wage premium over their competently trained peers, pinned by their firms' revenue; net of education costs, however, they accept a strictly lower payoff — the status discount that finances the system (Corollary 1). The premium decomposes exactly (gross premium equals excess tuition minus the status discount), so the status-financed component of elite pay is tuition reimbursement rather than productivity, and the decomposition yields the model's sharpest prediction: changes in elite tuition have zero incidence on elite salaries, being absorbed by the status discount. The waste is thus financed by identified parties at market prices; no friction, information asymmetry, or market power is involved. The same accounting yields a sharp possibility theorem: even *strictly less productive* elite firms survive whenever the status discount exceeds the workers' compensating differential plus the resources burned (Proposition 1) — though in that polar case the gross premium necessarily reverses, so the empirically relevant configuration is the one in which elite firms are more productive per firm and their managers visibly better paid.

Second, as the taste for status spreads through the population, aggregate output falls monotonically (Proposition 2). What matters is not productivity per firm but output *per unit of resources consumed*: in our central configuration elite firms produce strictly more output than competent ones, look superior in any firm-level comparison, pay their managers more — and still drag aggregate output down, because the elite school system burns resources that firm accounts never show.

Third (and centrally) every such equilibrium lies in the core of the economy and is Pareto optimal (Theorem 1). No entrant offering "status without waste," no coalition of workers and investors, and no policymaker restricted to lump-sum instruments can achieve a Pareto improvement. The market has nothing to correct: falling measured output is the equilibrium price of status consumption. The popular syllogism "markets work, therefore materially wasteful institutions cannot survive competition" commits a non sequitur, and the point of this paper is to locate it precisely.

The efficiency of the baseline is not a celebration; it is a diagnosis. It locates exactly where a genuine market failure must live if there is one: not in the waste, which is priced, but in the *relativity* of status. Section 7 makes status positional (valuable because others lack it) so that each entrant into the elite dilutes the status of every incumbent. This economy-wide externality lies outside the assumptions of the clubs welfare theorems: membership prices internalize composition effects *within* a group, but eliteness is a characteristic of a class that spans all elite groups. The positional equilibrium exists, is unique, and self-limits (dilution shrinks the elite), but it is no longer Pareto optimal and no longer in the core (Propositions 3 and 4): there is over-entry into elite institutions, and a Pigouvian tax on elite memberships raises welfare *and* output (Proposition 5). The contrast between Theorem 1 and Proposition 4 is the paper's answer to when elite waste is and is not a policy problem: absolute status is efficient consumption; positional status is a rat race.

Section 8 sets the economy in motion. Each generation inherits the resources its parents saved, plays the club economy within its lifetime, and bequeaths; the static model is the period economy and its comparative statics become the transition map. A single condition (the one already governing the static output decline) delivers two further results. Status is a *luxury* by theorem: because the elite sector's waste is a fixed resource cost while wages scale with wealth, richer economies support larger elites (Proposition 7). And the elite sector drags on accumulation: growth finances the status spending that arrests it, and the economy converges to a steady state permanently poorer than its status-free counterfactual (Proposition 8) — with no collapse, because impoverishment itself prices the elite out of existence at low wealth. The path involves no error and no myopia: it is common knowledge, and no atomistic dynasty could or would deviate. Rational societies really do let it happen.

Three features distinguish the exercise from prior work, which Section 2 reviews in detail. Status here is *produced and sold* by institutions with real resource costs and competitively priced as a club membership, rather than attached to occupations, allocated by norms, sold by a monopolist, or traded as an abstract good in fixed supply. The persistence result is at the level of the *firm*, with the financing of elite inefficiency traced to identified parties at market prices. And the welfare analysis is disciplined by the clubs welfare theorems themselves, so that the line between efficient status consumption and a genuine rat race is a theorem, not a stance.

The paper proceeds as follows. Section 2 positions the contribution. Section 3 presents the general framework and Section 4 our economy. Section 5 characterizes equilibrium, the compensation decomposition, and the no-market-correction theorem, illustrated numerically in Section 6. Section 7 develops the positional extension and Section 8 the overlapping-generations dynamics. Section 9 collects the model's empirical content, Section 10 discusses interpretation and extensions, and Section 11 concludes. All derivations are gathered in the appendices and are verified symbolically in the paper's companion code.

2 Related Literature

Markets for status. The idea that status itself can be traded has an established lineage. Becker et al. (2005) price status as a divisible good in fixed supply traded in a hedonic market alongside consumption; their central objects are the induced income distribution and the complementarity between status and consumption. We differ in making the supply side explicit: status here is membership in clubs (schools and the firms their graduates run) produced by institutions with real resource costs and priced by the membership prices of Ellickson et al. (1999), with a full production economy attached, so that the resource cost of status and its incidence across workers, managers, and schools are equilibrium objects rather than primitives. Cole et al. (1992) generate status-driven distortions from self-enforcing matching norms, in which rank governs access to nonmarket goods; our status is marketed rather than normed, which yields prices, entry margins, and comparative statics that norms models do not deliver, at the cost of taking the taste for membership as a primitive. A third tradition studies a single designer selling status: contest categories in Moldovanu et al. (2007), exclusive groups with peer effects in Board (2009), pure signals of type in Rayo (2013), and the fashion cycles of Pesendorfer (1995), in which an intrinsically useless good profitably sorts its buyers. Here, by contrast, the supply of status is competitive (any potential group type is priced) and embedded in general equilibrium, which is what allows welfare statements about the whole economy rather than about a designer's profit. Conspicuous-consumption signaling (Bagwell and Bernheim, 1996) shares our subject but not our mechanism: our economy has no hidden types, so the persistence of elite waste cannot be attributed to information.

Status and allocation. Fershtman et al. (1996) show occupational status concerns distort the allocation of talent in growth. Our persistence result is at the level of the *firm*: identifiable, resource-inefficient organizations survive (even output-inferior ones can) because their control rights are bundled with status consumption. The rat-race tradition (Akerlof 1976; Hopkins and Kornienko 2004; Frank 1985; Hirsch 1976) delivers genuine inefficiency from relative comparison; in our baseline there is no externality and hence no rat race — that distinction is the point of Theorem 1, and Section 7 shows exactly what must be added to overturn it, using the widespread-externalities equilibrium concept of Noguchi and Zame (2006). Kim et al. (2024) provide a recent quantitative treatment of education-status externalities; we provide the institutional theory.

Education and elite institutions. In Spence (1973) education is privately valuable because it is informative about hidden types. Our elite school conveys no information (there are no hidden types anywhere in the model) and its product is the membership itself; the mechanism would survive full information, which cleanly separates it from signaling. This places us closest to the reputation-and-selection view of schooling of MacLeod and Urquiola (2019), in which schools are valued for whom they admit rather than what they teach, and to the empirical literature finding that elite institutions' value operates through access and affiliation rather than measured skill production: near-zero average selectivity premia (Dale and Krueger, 2002), elite-program returns concentrated on the already-connected (Zimmerman, 2019), social-club premia exceeding academic premia (Michelman et al., 2022), admission effects operating through gateways to prestigious firms (Chetty et al., 2026), and hiring at elite professional-service firms screening on pedigree (Rivera, 2015). We treat the zero-teaching elite school as a polar case; Arteaga (2018) shows curricular content matters even at a top university, and nothing in our results requires the polar case (Remark 1).

The clubs framework. Our machinery descends from Buchanan (1965) through the competitive general-equilibrium formulation of Ellickson et al. (1999, 2001, 2006); Scotchmer (2002) surveys the field, Cole and Prescott (1997) provides the lottery formulation, and Allouch and Wooders (2008) relax the bounds on club sizes. Applied uses of this machinery are rare: the main exceptions are Caucutt (2001, 2002), who compute club equilibria of school economies with peer effects, and the firms-as-clubs analyses of Prescott and Townsend (2006) and Zame (2007), the latter establishing that with strategic play inside firms, equilibria may be Pareto ranked. To our knowledge the present paper is the first to use the framework to price status and the first to embed a widespread externality in it; we are also not aware of prior work studying transition dynamics whose period-by-period allocations are club equilibria, though computed club economies of schooling (Caucutt, 2001, 2002) come closest.

Dynamics. The rent-seeking-and-decline tradition (Murphy et al. 1991; Baumol 1990; Acemoglu 1995) diverts talent to unproductive activity through distorted rewards; our drag operates instead through competitively priced status consumption, and its strength is disciplined by the same condition that governs the static equilibrium. Where Hassler and Rodríguez Mora (2000) and Cole et al. (1992) sustain aristocratic regimes through information or norms, our elite is sustained by prices — and where the leisure-loving aristocracies of Doepke and Zilibotti (2008) are competed away at industrialization, ours survive precisely because their inefficiency is financed by willing status buyers rather than borne by the firms' bottom line. The memberships tradition in inequality dynamics (Durlauf 1996; Galor and Zeira 1993) generates traps from productive group memberships and credit constraints; our membership is a pure positional good, and our poverty dynamics run top-down (wealth creates the elite that erodes wealth) rather than bottom-up. Finally, the pattern our dynamics formalize, prosperity breeding an expanding elite that strains the society supporting it, is the "elite overproduction" of Turchin (2023). That literature is historical and demographic; to our knowledge, no formal treatment with optimizing agents and market-clearing prices exists, and Propositions 7 and 8 may be read as one.

3 The Framework

We adopt the general equilibrium model of groups of Ellickson et al. (2006), which extends Ellickson et al. (1999) to allow groups to produce outputs and memberships to carry acquired characteristics. This section states the primitives, explains what each is intended to represent, and records the two theorems we use; the reader is referred to the source for proofs and for the framework's full generality.

Notational conventions. Throughout the paper, lowercase Latin and Greek letters denote parameters, functions, and elements of sets; capital Latin and Greek letters denote sets; and calligraphic letters denote collections of sets (the only instance below is a σ-algebra). Blackboard letters denote the standard number systems: $\mathbb{R}$ is the real line, $\mathbb{R}^N$ its N-fold Cartesian product, $\mathbb{R}_+$ the nonnegative reals, and $\mathbb{Z}_+ = \{0, 1, 2, \dots\}$ the nonnegative integers. Two standard exceptions are retained: distribution functions are written with the conventional capital F, and a bar over a lowercase letter ($\bar{w}$, $\bar{y}$) denotes an economy-wide aggregate of the corresponding individual quantity. Every symbol is defined where it first appears. Two workhorse letters serve double duty across distant contexts where no ambiguity can arise: s is an organizational characteristic in the group-type notation and the taste-scale parameter of the numerical examples, and t is the membership tax of Section 7 and the time index of Section 8.

3.1 Commodities, groups, and memberships

There are $N \geq 1$ perfectly divisible, publicly traded private goods; a bundle of them is a vector in $\mathbb{R}^N$. Alongside these ordinary commodities, the framework's distinctive objects are *groups*: collections of individuals associated for some purpose — in our application, schools and firms.

Two exogenous finite sets describe the raw material from which groups are built. Ω is the set of *membership characteristics*, with typical element ω. A membership characteristic is any attribute of a person that matters to the group she joins or to the activity conducted there: a role (student, laborer), a qualification (holding a particular degree), or a position (manager). Characteristics are observable and contractible, and (importantly for what follows) they can be *acquired*: attending a school confers a characteristic that a firm requires. Γ is the set of *organizational characteristics*, with typical element γ. An organizational characteristic describes everything about a group that is not its membership roster or its market transactions: its activity, internal organization, or institutional identity. In our application Ω has four elements (student, competent manager, elite manager, laborer) and Γ has four (serious school, elite school, competent firm, elite firm); both sets are primitives of the model, fixed throughout.

A *group type* is a triple (π, γ, y) consisting of: a *profile* $\pi : \Omega \to \mathbb{Z}_+$, an exogenously fixed function specifying the integer number $\pi(\omega)$ of members with characteristic ω that a group of this type requires, so that $\sum_{\omega \in \Omega} \pi(\omega)$ is the group's total size; an organizational characteristic $\gamma \in \Gamma$; and an *input–output vector* $y \in \mathbb{R}^N$, whose negative components are the quantities of private goods the group uses up and whose positive components are the quantities it produces. A firm, for example, is a group type whose profile demands one manager and two laborers and whose input–output vector converts raw materials into product; a school is a group type whose profile demands one student and whose input–output vector uses up resources and produces no traded good. The economy's group technology is an exogenous finite set G of group types. Groups of any type in G may form in any nonnegative mass; which types form, and in what mass, is determined in equilibrium.

A *membership* is an opening in a group: formally, a pair $m = (\omega, (\pi, \gamma, y))$ with $(\pi, \gamma, y) \in G$ and $\pi(\omega) \geq 1$ — a seat for a member with characteristic ω in a group of type (π, γ, y). M denotes the set of memberships; it is finite because Ω and G are. Agents may hold several memberships at once (the same person can be a student in one group and a manager in another), so an agent's choice object is a *list*: a function $\ell : M \to \mathbb{Z}_+$ recording how many memberships of each kind she holds. Λ denotes the set of lists, and $0 \in \Lambda$ denotes the empty list (no memberships).

3.2 Agents

The set of agents is a nonatomic finite measure space $(A, \mathcal{F}, \iota)$: A is the set of agents, $\mathcal{F}$ is a σ-algebra of subsets of A (the collection of coalitions to which a size can be assigned) and ι is a nonatomic measure with $\iota(A) < \infty$, so that $\iota(B)$ is the mass of coalition $B \in \mathcal{F}$ and no single agent carries positive mass. The continuum formalizes perfect competition: each agent is negligible and takes prices as given, while groups of small finite size can still form in positive mass — the technical feature that makes competitive pricing of memberships coherent.

Each agent $a \in A$ is described by three objects. Her *consumption set* $X_a \subset \mathbb{R}^N_+ \times \Lambda$ collects the pairs of goods bundles and lists she is capable of choosing; it is closed, allows free upward disposal in private goods, and bounds the number of memberships on any list by a uniform integer $\bar{\ell}$. The consumption set is where qualifications live: a list that includes a managerial membership belongs to X_a only if it also includes the schooling membership that confers the managerial qualification. Her *endowment* $(e_a, 0) \in X_a$ consists of a bundle $e_a \in \mathbb{R}^N_+$ of private goods and the empty list: nobody is born holding a membership. Her *utility function* $u_a : X_a \to \mathbb{R}$ is continuous and strictly increasing in private goods for each fixed list; no assumption is placed on how utility varies across lists, which is precisely what allows preferences over group affiliation (including a taste for elite status) to enter unrestricted. The economy is the mapping $a \mapsto (X_a, e_a, u_a)$: the consumption-set correspondence is measurable, the endowment mapping integrable, and utility jointly measurable in its arguments.

3.3 States, feasibility, and equilibrium

A *state* is a measurable assignment $(x, \ell) : A \to \mathbb{R}^N_+ \times \Lambda$ of a goods bundle x_a and a list ℓ_a to each agent. A state is *feasible* if three requirements hold. First, *individual feasibility*: $(x_a, \ell_a) \in X_a$ for almost every $a \in A$. Second, *consistency*: the memberships chosen across the population can actually be assembled into groups. Formally, for every group type $(\pi, \gamma, y) \in G$ there is a scalar $\nu(\pi, \gamma, y) \geq 0$ (the mass of groups of that type that form) such that

$$\int_A \ell_a\big(\omega, (\pi, \gamma, y)\big)\, d\iota(a) \;=\; \nu(\pi, \gamma, y)\, \pi(\omega) \qquad \text{for every } \omega \in \Omega :$$

the population's choices of the various seats in a group type must come in exactly the proportions its profile requires, so that groups form with no seat empty and no member unmatched. Third, *material balance*:

$$\int_A x_a\, d\iota(a) \;\leq\; \int_A e_a\, d\iota(a) \;+ \sum_{(\pi,\gamma,y)\in G} \nu(\pi, \gamma, y)\, y,$$

aggregate consumption cannot exceed aggregate endowment plus the net output of all the groups that form. A state is *feasible for a coalition* $B \in \mathcal{F}$ if the same three conditions hold with all integrals taken over B; this is the notion under which a coalition can block, and hence under which the core is defined.

Both goods and memberships are priced. A price system is a pair (p, q) with $p \in \mathbb{R}^N_+$ the prices of the private goods and $q \in \mathbb{R}^M$ a vector assigning a price $q(m)$ to each of the finitely many membership kinds $m \in M$. Membership prices are the framework's payroll and fee schedule rolled into one object: a positive $q(m)$ is a payment the member makes to the group (tuition), and a negative $q(m)$ is a payment the member receives (a wage). A *group equilibrium* is a price system (p, q) with $p \neq 0$ and a feasible state (x, ℓ) such that:

1. **Budget balance for group types:** for each $(\pi, \gamma, y) \in G$, $\sum_{\omega\in\Omega} \pi(\omega)\, q(\omega, (\pi, \gamma, y)) + p \cdot y = 0$. Every group breaks even: the membership payments its members make and receive, plus the market value of its net output, sum to zero — there are no profits to distribute and no owners required.

2. **Budget feasibility:** for almost all a, $(p, q) \cdot (x_a, \ell_a) \leq p \cdot e_a$. Each agent's spending on goods and memberships, net of the wages her memberships pay, is covered by the value of her endowment.

3. **Optimization:** for almost all a, any choice in X_a strictly preferred to (x_a, ℓ_a) costs strictly more than $p \cdot e_a$. No agent can afford anything she would rather have.

Two results from Ellickson et al. (2006) are used below. Their Theorem 1: *every group equilibrium state belongs to the core, and in particular is Pareto optimal; if endowments are desirable, it belongs to the strong core and is strongly Pareto optimal.* Their Theorem 2 provides existence under group irreducibility, desirable endowments, and uniformly bounded endowments; our analysis will not need it, because we exhibit equilibria in closed form.

4 An Economy with a Market for Elite Status

4.1 Goods, group types, and memberships

There are two private goods: an input good y_1 ("resources," the numéraire) and a consumption good y_2 with price p_2. The membership characteristics are $\Omega = \{\omega_s, \omega_m, \omega_e, \omega_l\}$: ω_s is being a student, ω_m holding a competent managerial qualification, ω_e holding an elite managerial qualification, and ω_l supplying labor, which requires no qualification. The organizational characteristics are $\Gamma = \{s, s^*, f, f^*\}$: serious school, elite school, competent firm, elite firm. There are four group types:

$$
\begin{aligned}
g_1 &= \big(\pi_1;\ s;\ (-r, 0)\big) && \text{serious school: educates one student at resource cost } r > 0,\\
g_2 &= \big(\pi_1;\ s^*;\ (-\tau r, 0)\big) && \text{elite school: } \tau \geq 1,\\
g_3 &= \big(\pi_2;\ f;\ (-c, \phi)\big) && \text{competent firm: one manager, two laborers,}\\
g_4 &= \big(\pi_3;\ f^*;\ (-\beta c, \alpha\phi)\big) && \text{elite firm: } \beta \geq 1,\ \alpha > 0.
\end{aligned}
$$

Here π_1 is the profile demanding one student and nothing else ($\pi_1(\omega_s) = 1$, zero elsewhere); π_2 demands one competent manager and two laborers ($\pi_2(\omega_m) = 1$, $\pi_2(\omega_l) = 2$); and π_3 demands one elite manager and two laborers ($\pi_3(\omega_e) = 1$, $\pi_3(\omega_l) = 2$). Thus $G = \{g_1, g_2, g_3, g_4\}$, and the goods space has $N = 2$. The parameters $r > 0$ and $c > 0$ are the resource costs of operating a serious school (per student) and a competent firm; $\phi > 0$ is the competent firm's output. The parameter τ measures the resources elite schools devote to the production of status rather than skill; β measures the excess resource use of elite-managed firms; α their output multiplier. The configuration of central interest is $\tau > 1$, $\beta \geq 1$, and α *above* one but below the resource-efficiency threshold identified in Proposition 2: elite firms out-produce competent ones firm-for-firm, yet the elite complex (schools included) consumes disproportionately more resources than it returns. The polar case $\alpha \leq 1$, in which even output-inferior elite firms survive, is characterized in Proposition 1.

The resulting membership set is $M = \{m_1, m_2, m_{3.1}, m_{3.2}, m_{4.1}, m_{4.2}\}$, where $m_1 = (\omega_s, g_1)$ and $m_2 = (\omega_s, g_2)$ are student seats at the two schools, $m_{3.1} = (\omega_m, g_3)$ and $m_{4.1} = (\omega_e, g_4)$ are the managerial seats at the two firm types, and $m_{3.2} = (\omega_l, g_3)$ and $m_{4.2} = (\omega_l, g_4)$ are laborer seats.

4.2 Agents

Agents form a continuum of measure one. Each is endowed with $e = (k, 0)$, $k > 0$. Agents may attend at most one school and hold at most one job. The qualification ω_m is acquired at a serious school and ω_e at an elite school, encoded in consumption sets as in Section 3: a list with $\ell(m_{3.1}) \geq 1$ is feasible only if $\ell(m_1) \geq 1$, and $\ell(m_{4.1}) \geq 1$ only if $\ell(m_2) \geq 1$. The characteristic ω_l requires no schooling, so laborer seats are open to all.

Agents differ in a single taste parameter $\eta \geq 0$, distributed according to an atomless distribution function F, strictly increasing on its support $[0, \bar{\eta})$ with $\bar{\eta} \leq \infty$, differentiable with derivative F' where needed below, and with finite mean. When we consider shifts of the taste distribution, a *FOSD shift* means a distribution $\tilde{F}$ with $\tilde{F} \leq F$ pointwise, and we call the shift *strict at* η' if $\tilde{F}(\eta') < F(\eta')$. Preferences are

$$u(x, \ell; \eta) \;=\; d(\ell; \eta)\, h(x_1, x_2), \qquad h(x_1, x_2) = \sqrt{x_1 x_2}, \tag{1}$$

where d is built multiplicatively from the list: attending any school contributes a factor $\delta_s \in (0, 1)$; working as a manager a factor $\delta_m \in (0, 1)$; working as a laborer a factor $\delta_l \in (0, 1)$; holding the elite-manager membership $m_{4.1}$ contributes the additional factor $(1 + \eta)$ (this is the utility of elite status) and holding the elite-firm laborer membership $m_{4.2}$ the factor $(1 - \chi)$, $\chi \in [0, 1)$, reflecting distaste for working under elite management. Thus, for the five undominated vocational choices,

$$d = \begin{cases} 1 & \text{leisure (empty list),} \\ \delta_s \delta_m (1 + \eta) & \text{elite manager } (\ell(m_2) = \ell(m_{4.1}) = 1), \\ \delta_s \delta_m & \text{competent manager } (\ell(m_1) = \ell(m_{3.1}) = 1), \\ \delta_l & \text{laborer, competent firm } (\ell(m_{3.2}) = 1), \\ \delta_l (1 - \chi) & \text{laborer, elite firm } (\ell(m_{4.2}) = 1). \end{cases}$$

The consumption sets already exclude a second school or a second job; the remaining feasible lists (school without subsequent employment, schooling combined with labor) are strictly dominated at any prices arising in equilibrium, as Lemma 2 in Appendix A verifies. Note that status attaches to the *elite-manager membership*: the elite school is the gate, not the prize.

It is convenient to write

$$\mu \equiv \frac{1}{\delta_s \delta_m} > 1, \qquad \lambda \equiv \frac{1}{\delta_l} > 1, \qquad \lambda^e \equiv \frac{\lambda}{1-\chi} \geq \lambda.$$

Remark 1. *The elite school teaches nothing and the elite firm wastes resources by assumption; these are polar cases chosen for transparency, not necessity. All results below depend only on the composite parameters* (τ, β, α)*, so a reader who prefers elite schools that teach something (but less per dollar) may set* τ *accordingly without change to the analysis.*

5 Equilibrium

Because h is homothetic, an agent with net income w (endowment value net of membership prices) demands $x_1 = w/2$, $x_2 = w/(2p_2)$ and attains indirect utility $d\,w/(2\sqrt{p_2})$. All occupational comparisons are therefore rankings of $d \cdot w$, independent of p_2.

5.1 Characterization

Occupational indifference for the (atomless) population pins down net incomes. Since leisure yields $d \cdot w = k$, participation of laborers and competent managers requires

$$w_M = \mu k, \qquad w_L = \lambda k, \qquad w_{L^*} = \lambda^e k, \tag{2}$$

and the elite-manager income satisfies, for a *marginal* type η^*,

$$w_E = \frac{\mu k}{1+\eta^*}, \tag{3}$$

with all types $\eta > \eta^*$ strictly preferring the elite path and all $\eta < \eta^*$ strictly avoiding it. Membership prices follow from the definition of net income ($w = k - \text{fees} + \text{wages}$) and budget balance of the schools:

$$q_1 = r, \quad q_2 = \tau r, \quad q_{3.1} = k - r - \mu k, \quad q_{3.2} = k - \lambda k, \quad q_{4.1} = k - \tau r - w_E, \quad q_{4.2} = k - \lambda^e k. \tag{4}$$

Budget balance for competent firms determines the output price,

$$p_2 \phi = c + r + (\mu + 2\lambda - 3)k \ > \ 0, \tag{5}$$

and budget balance for elite firms determines the cutoff. Define

$$z \ \equiv \ \alpha\big[c + r + (\mu + 2\lambda - 3)k\big] \ - \ \beta c \ - \ \tau r \ + \ (3 - 2\lambda^e)k. \tag{6}$$

Then the elite-firm budget balances if and only if $w_E = z$, i.e.,

$$1 + \eta^* \ = \ \frac{\mu k}{z}. \tag{7}$$

The measure of elite managers is $m^* = 1 - F(\eta^*)$; consistency requires the elite sector to employ $2m^*$ laborers and the competent sector, of measure n firms, to employ n managers and $2n$ laborers. Market clearing in the input good,

$$\underbrace{\sigma_0 \frac{k}{2} + m^* \frac{w_E}{2} + 2m^* \frac{w_{L^*}}{2} + n \frac{w_M}{2} + 2n \frac{w_L}{2}}_{\text{consumption demand}} + \underbrace{m^*(\tau r + \beta c) + n(r + c)}_{\text{resource use}} \ = \ k, \tag{8}$$

with $\sigma_0 = 1 - 3m^* - 3n$ the leisure share, determines n linearly; clearing of the consumption good follows by Walras' law (Appendix A).

Assumption 1 (Interior configuration)**.** *Parameters are such that (i) $0 < z < \mu k$; (ii) $F(\eta^*) < 1$; (iii) the solution n of (8) is strictly positive; and (iv) $\sigma_0 > 0$: a positive measure of agents chooses the empty list.*

Part (i) puts the cutoff in the interior: if $z \le 0$, elite firms cannot break even even with unpaid managers and the elite sector is empty; if $z \ge \mu k$, every agent joins it. Parts (ii)–(iv) ensure all sectors coexist. The parameterization of Section 6 satisfies all four.

Lemma 1. *Under Assumption 1, the prices (4)–(5) and the state in which agents with $\eta > \eta^*$ sort into elite management, measures n, $2n$, and $2m^*$ of agents with $\eta < \eta^*$ sort into competent management, competent labor, and elite labor respectively, and the remainder choose leisure, constitute a group equilibrium. Every group equilibrium in which* *all six memberships are chosen by a positive measure of agents* and *a positive measure of agents holds the empty list has these prices, and its cutoff, sector sizes, and allocation are given by* (3)–(8)*. (The leisure margin anchors the price system: without it, the non-elite occupations need only deliver a common utility level, and the wage normalization* (2) *is not forced.)*

The proof (Appendix A) verifies budget balance, budget feasibility, consistency, and (using Lemma 2) optimization over every feasible list.

Two features deserve notice. First, elite managers *accept a status discount* in net terms: $w_E < w_M$ in every interior equilibrium, with the discount $\mu k\, \eta^*/(1+\eta^*)$ increasing in the equilibrium cutoff. In membership-price terms, the elite *career bundle* is always the more expensive one: $q_2 + q_{4.1} > q_1 + q_{3.1}$, tuition and job taken together. Whether the job taken alone pays more (whether observed elite *salaries* exceed competent ones) is a separate question, settled by Corollary 1 below. Second, laborers in elite firms are paid a compensating differential, $w_{L^*} > w_L$: as Ellickson et al. (2006, p. 151) anticipate, firms offering uncongenial conditions must pay more.

5.2 Gross premia, net discounts: what compensation data show

The membership prices (4) separate what an elite manager is *paid* from what an elite career *nets*. The gross wage of an elite manager is $-q_{4.1} = w_E - k + \tau r$; of a competent manager, $-q_{3.1} = w_M - k + r$.

Corollary 1 (Compensation decomposition)**.** *In any interior equilibrium,*

$$\underbrace{(-q_{4.1}) - (-q_{3.1})}_{\textit{gross wage premium}} = \underbrace{(\tau - 1)r}_{\textit{excess tuition}} - \underbrace{(w_M - w_E)}_{\textit{status discount}}. \tag{9}$$

Hence: (i) elite managers earn a gross wage premium if and only if the status discount is smaller than the excess tuition; (ii) a strict gross premium implies $\alpha > 1$*, by the equilibrium identity* $p_2\phi(1-\alpha) = (w_M - w_E) - 2(w_{L^*} - w_L) - (\beta - 1)c - (\tau - 1)r$ *then forces* $\alpha > 1$*; (iii) in the polar case* $\alpha < 1$ *of Proposition 1 below, elite managers are paid less in gross terms as well.*

Corollary 1 is the model's answer to the most immediate empirical objection. Elite managers in the data are conspicuously well paid; in the model they are too, in the configuration $1 < \alpha < (\beta c + \tau r)/(c + r)$: their firms generate more revenue per firm and every firm-level comparison flatters them. What the wage carries, by the accounting identity (9), is tuition reimbursement net of the status discount (in the limit of a vanishing discount, $\eta^* \to 0$, it is exactly tuition reimbursement), and what the identity prices that data do not show is the net position: subtract the cost of the elite education, and the elite career pays strictly less than the competent one. The willingness to accept that worse net bargain is the status payment on which the entire elite complex is financed. Two scope remarks discipline the interpretation. The identity caps the pure status-financed premium at excess tuition, so observed elite premia that dwarf tuition differentials, as the causal estimates discussed in Section 9 imply, must carry additional components, scarcity rents and bundled ability foremost (Section 10); the decomposition then bounds the status component rather than the total. And the identity is an equilibrium accounting statement, not a comparative static: a change in elite tuition τr moves the status discount one for one and the gross wage not at all, a point Section 9 turns into the model's sharpest compensation prediction.

5.3 When do even unproductive elite firms survive?

Proposition 1 (Coexistence)**.** *Under Assumption 1, an equilibrium with an active elite sector of strictly lower output productivity ($\alpha < 1$) exists if and only if, at the equilibrium cutoff η^*,*

$$\underbrace{(w_M - w_E)}_{\textit{status discount}} > \underbrace{2\,(w_{L^*} - w_L)}_{\textit{compensating differential}} + \underbrace{(\beta - 1)c + (\tau - 1)r}_{\textit{resources burned}}. \tag{10}$$

The economics is an accounting identity made competitive: the elite firm's revenue shortfall relative to a competent firm must be financed within the group, and the only willing financier is the manager who values the membership itself. The status discount conceded by the marginal elite manager must cover both the premium demanded by the two laborers who would rather work elsewhere and the resources burned by the elite school and the elite firm's own inefficiency. No cross-subsidy from outside the group is possible (budget balance for group types forbids it) so the entire institution of wasteful eliteness is self-financing, at market prices, out of the demand for status.

5.4 The spread of status preferences and the decline of output

Aggregate output of the consumption good is $\bar{y} = \phi\,(n + \alpha m^*)$. Consider a first-order stochastically dominating shift of the taste distribution F — more people care more about status. By (7) the cutoff η^* depends only on prices and technology, *not* on F; hence the elite sector expands mechanically, $m^* = 1 - F(\eta^*)$ rising with the shift.

Proposition 2 (Efficient decline)**.** *Under Assumption 1, a FOSD shift of F that is strict at the cutoff η^* strictly increases the mass of elite managers and firms, and strictly decreases aggregate output if and only if*

$$\frac{\alpha\phi}{\beta c + \tau r} < \frac{\phi}{c + r}, \tag{11}$$

i.e., if and only if the elite sector produces less output per unit of resources than the competent sector. In particular, output declines whenever $\alpha \leq 1$ and $(\tau - 1)r + (\beta - 1)c > 0$; but it also declines for a range of $\alpha > 1$.

Condition (11) is strictly weaker than $\alpha < 1$: an elite sector whose firms produce *more* output than competent firms still drags aggregate output down if its schools and firms consume disproportionately more resources. How this drain surfaces in measured statistics depends on accounting conventions — Section 9 takes up the mapping.

5.5 The market cannot correct it

Theorem 1 (No market correction)**.** *Every equilibrium of Lemma 1 belongs to the core of the economy and is strongly Pareto optimal. Consequently: (i) no feasible reallocation — including any that shuts the elite sector, redirects its resources, or introduces any group type in G in any proportion — makes a positive measure of agents strictly better off while leaving almost every agent at least as well off; (ii) no coalition of agents, of any size, can profitably secede with any configuration of schools and firms; (iii) the conclusion is unchanged as the taste for status spreads and output falls per Proposition 2.*

Proof. The state is a group equilibrium (Lemma 1), so membership in the core and Pareto optimality follow from Theorem 1 of Ellickson et al. (2006). Strong Pareto optimality follows from the same theorem once endowments are verified to be desirable, which is done in Appendix B. □

Theorem 1 is the paper's central claim, and it is worth stating what it does and does not say. It does not say the elite sector is harmless: output falls, resources are burned, and the burn is real. It says the situation is not a market failure. The waste is voluntarily financed by the people who consume the status it produces, at prices that fully internalize its cost within each group. Competition has no lever: an entrant offering "elite status without the waste" would have to offer the elite-manager membership at its equilibrium price, and the equilibrium already prices every potential group type in G.

The familiar inference (markets discipline waste, elite waste persists, therefore markets must be impeded) fails at its first step. Markets discipline waste that is unpriced; status is priced.

Three caveats delimit the result. First, the feasible set takes the group-type list G as given; whether a status-conferring group type without resource burn could exist is a question about what makes status credible. Section 7 answers it: when status derives from scarcity, entry self-dilutes, and exclusivity needs no artificial protection. Second, if status is valued *because others lack it*, utility depends on the economy-wide pattern of memberships — an externality outside the present framework, under which the welfare theorem genuinely fails; that is the subject of Section 7. Third, welfare here is welfarist: status utility counts. Section 10 returns to this.

6 Numerical Example

Let $k = 2$, $r = c = 1$, $\tau = 4$, $\beta = 1.25$, $\alpha = 1.25$, $\phi = 6$, $\delta_s = \delta_m = \frac{1}{2}$ (so $\mu = 4$), $\delta_l = \frac{1}{2}$ (so $\lambda = 2$), $\chi = 0.2$ (so $\lambda^e = 2.5$), and let η be exponentially distributed with mean $s = 0.1$. Elite schools cost four times what serious schools cost. Elite firms are competent firms scaled up by exactly 25% (both input and output) so no firm-level productivity comparison distinguishes them; the entire drain is the school system's burn, $(\tau - 1)r = 3$ per elite manager trained. The resource condition (11) holds with room to spare: $\alpha(c + r) = 2.5 < \beta c + \tau r = 5.25$.

Then $p_2 = 2$, $w_E = z = 5.75$, $\eta^* = 9/23 \approx 0.391$, and:

	elite manager	competent manager	laborer (elite firm)	laborer (comp. firm)
gross wage $-q$	7.75	7	3	2
tuition paid	4	1	—	—
net income w	5.75	8	5	4

The example displays Corollary 1 in the empirically relevant regime: elite managers earn a visible salary premium of 10.7% (7.75 against 7), which is exactly excess tuition minus the status discount ($3-2.25 = 0.75$), while their net position is 28% *below* their competent peers' (5.75 against 8). The elite sector employs $3m^* \approx 6.0\%$ of the population ($m^* \approx 0.020$), the competent sector $3n \approx 34.2\%$ ($n \approx 0.114$), and $\sigma_0 \approx 59.8\%$ take leisure. Aggregate output is $\bar{y} \approx 0.834$ and the resources burned are $m^*[(\beta-1)c+(\tau-1)r] \approx 0.065$. For the polar regime of Proposition 1, set $\tau = 2$, $\alpha = 0.9$, $s = 0.3$: coexistence condition

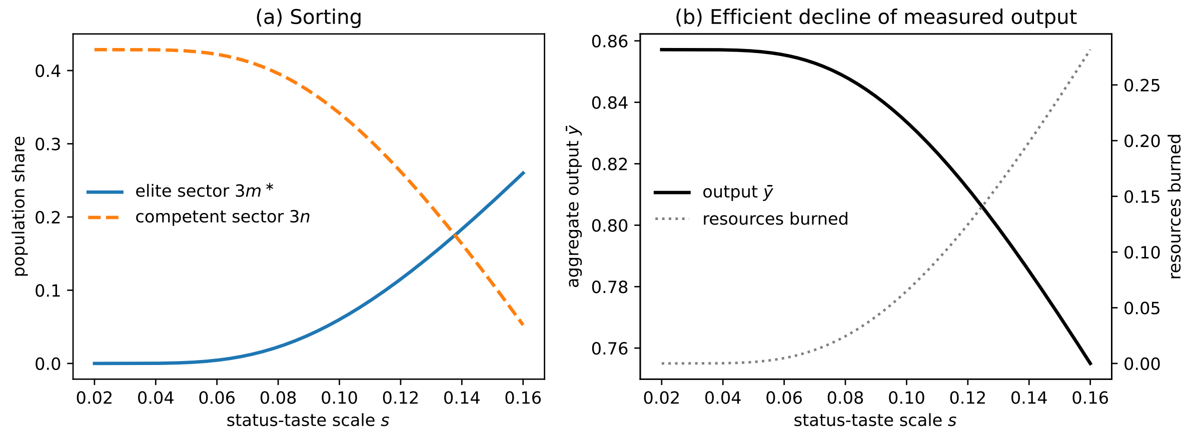

Figure 1: Comparative statics in the status-taste scale s ($\eta \sim$ Exponential, mean s). Panel (a): population shares of the elite and competent sectors. Panel (b): aggregate output falls and burned resources rise as status taste spreads; every equilibrium shown is Pareto optimal (Theorem 1). Parameters as in Section 6.

(10) holds ($4.45 > 2 + 1.25$) and, per Corollary 1(iii), elite managers are then paid less in gross terms as well (3.55 against 7).

Figure 1 traces the equilibrium as the mean s of the taste distribution rises — a FOSD shift. The elite sector expands, the competent sector contracts, and output falls monotonically, while by Theorem 1 every point on the path is in the core.

7 Positional Status and the Failure of the Welfare Theorem

Theorem 1 rests on status being an *absolute* amenity: the elite-manager membership yields utility $(1 + \eta)$ regardless of how many others hold it. But much of what makes status status is scarcity (Hirsch, 1976; Frank, 1985): a credential everyone holds confers distinction on no one. This section makes status positional and shows that everything in Theorem 1 breaks — selectively, and in an instructive way.

7.1 Positional preferences and equilibrium

Let m denote the aggregate measure of agents holding the elite-manager membership $m_{4.1}$, and replace the status factor $(1+\eta)$ in the utility specification of Section 4 with

$$1 + \eta\, v(m), \tag{12}$$

where $v : [0,1] \to (0,1]$ is continuously differentiable, strictly decreasing, with $v(0) = 1$: the value of eliteness falls as eliteness spreads. Each agent is negligible and takes m as given. (The scalar mass m is unsubscripted; membership labels $m_1, m_{3.1}, \ldots$ always carry subscripts, so no confusion should arise, and in equilibrium m coincides with the elite-manager mass m^* of Section 5.) Because utility now depends on *other agents'* memberships, preferences violate the assumptions of Ellickson et al. (2006): this is a widespread externality in the sense of Noguchi and Zame (2006). We accordingly define a *positional group equilibrium* as prices (p, q) and a feasible state (x, ℓ) satisfying the three conditions of Section 3, where optimization is with respect to preferences (12) evaluated at the aggregate elite mass m induced by the state itself.

Crucially, the externality operates at the level of the *class* of elites (the holders of $m_{4.1}$ across all elite firms) not at the level of any single group. Membership prices internalize composition effects within a group; they have no instrument that spans groups. This is precisely the gap in the price system that the welfare theorem needs closed, and it is why the failure below is not a defect of the clubs framework but a fact the framework makes visible.

7.2 Existence, uniqueness, and self-limiting eliteness

The wages (2), the output price (5), and the elite-firm break-even income $w_E = z$ are untouched by the externality. The cutoff condition becomes $(1 + \eta^* v(m))\, \delta_s \delta_m\, w_E = k$, i.e.,

$$\eta^*(m) \;=\; \frac{\eta_0^*}{v(m)}, \tag{13}$$

where $\eta_0^* = \mu k/z - 1$ is the cutoff of Section 5. Consistency of the conjectured mass with sorting requires m to be a fixed point of

$$\rho(m) \;\equiv\; 1 - F\left(\frac{\eta_0^*}{v(m)}\right). \tag{14}$$

Proposition 3 (Positional equilibrium)**.** *Under Assumption 1 (maintained at the resulting equilibrium configuration), a positional group equilibrium exists and its elite mass is unique: ρ is continuous and weakly decreasing — strictly wherever positive — so $\rho(m)-m$ is strictly decreasing and has exactly one root m^*, which is strictly positive whenever the support of F extends beyond $\eta_0^*/v(0)$. Moreover: (i) if $m^* > 0$ then $m^* < m_0 \equiv 1 - F(\eta_0^*)$ — dilution self-limits the elite; (ii) a FOSD shift of F that is strict at the positional cutoff raises m^*; if in addition the shift's vertical size at the positional cutoff $\eta_0^*/v(m^*)$ is weakly smaller than at η_0^* — as holds for the exponential scale family used in our examples — then m^* rises by strictly less than m_0, so positional concerns dampen the expansion of Proposition 2; (iii) all prices, and the equilibrium values of p_2, w_E, n, and $\bar{y}$ given m^*, are as in Section 5.*

The qualification in (ii) is substantive, not decorative: because the positional cutoff $\eta_0^*/v(m^*)$ exceeds η_0^*, a taste shift concentrated between the two cutoffs would move the positional mass and not the baseline mass, reversing the comparison; damping is a property of shifts that do not favor precisely the diluted region.

Part (i) already delivers a result the baseline cannot: eliteness is self-protecting. Entry of new elite institutions dilutes the very good they sell, so the market itself polices exclusivity — no entrant can arbitrage status, because relative position cannot be replicated by replication.

7.3 Failure of optimality and of core equivalence

Proposition 4 (Inefficiency)**.** *If v is strictly decreasing, the positional group equilibrium is not Pareto optimal, and its state does not belong to the core.*

The proof (Appendix C) is constructive and exploits the equilibrium's own structure. Every non-elite agent (laborers, competent managers, leisure-takers) is at reservation utility and indifferent among their activities; so is every marginal elite. The planner therefore shuts a small mass dm of elite triples and opens competent triples in the ratio that restores material balance in both goods — a reallocation that is *exactly* feasible because both adjustment directions have zero value at equilibrium prices. The released laborers and the redeployed leisure-takers are indifferent; the shut firms' managers lose only their near-marginal status surplus, an amount of second order in dm that the remaining elites' first-order de-dilution gain strictly covers through explicit compensation. But $v(m)$ rises, so every inframarginal elite is strictly better off: a Pareto improvement, executable by the grand coalition, so the state is blocked and core equivalence fails. The contrast with

Theorem 1 is exact: there, shutting elite firms had to take status away from people who had paid its price; here, shrinking the elite *creates* status out of scarcity.

7.4 Over-entry and the membership tax

At the laissez-faire equilibrium the marginal entrant weighs the private value of elite membership against its price, but ignores the dilution imposed on every inframarginal elite. The planner's cutoff condition (Appendix C) equates the marginal type's surplus to the aggregate dilution wedge

$$t^* \;=\; |v'(m^*)| \, \frac{w_E}{1+\eta^* v(m^*)} \int_{\eta \geq \eta^*} \eta \, dF(\eta), \tag{15}$$

expressed in units of the marginal entrant's income. Since the equilibrium sets the marginal surplus to zero while the wedge is strictly positive, there is over-entry.

Proposition 5 (Membership tax)**.** *Let a tax t be levied on the elite-manager membership, with revenue rebated as a uniform lump sum. (i) At any fixed rebate level, the fixed-point map remains continuous and strictly decreasing in the elite mass, so the equilibrium elite mass is unique given the rebate and is strictly decreasing in t. (ii) The laissez-faire equilibrium exhibits over-entry: its cutoff sets the marginal entrant's surplus to zero, whereas every Pareto-optimal cutoff allocation requires that surplus to equal the strictly positive wedge* (15)*. (iii) Along any equilibrium selection in which the elite mass falls with the tax, condition* (11) *implies that aggregate output rises with the tax.*

The propositions state what is proven; the example shows that the hypotheses are not vacuous. There, the joint fixed point in the elite mass and the rebate is unique, the elite mass falls with the tax, utilitarian welfare is strictly increasing at $t = 0$, and the welfare-maximizing tax is interior; Section 7 reports the magnitudes. Part (iii) deserves emphasis because it reverses the baseline's policy content. Under absolute status, taxing the elite sector reduces welfare (it undoes priced, efficient consumption). Under positional status, the tax raises measured output *and* welfare simultaneously — there is no equity–efficiency trade-off, because the taxed activity was destroying value at the margin in both senses. The practical diagnostic offered by the theory is thus a question about preferences, not about waste: *is elite status valued in itself, or valued because others lack it?* Only in the second case is there anything for policy to correct.

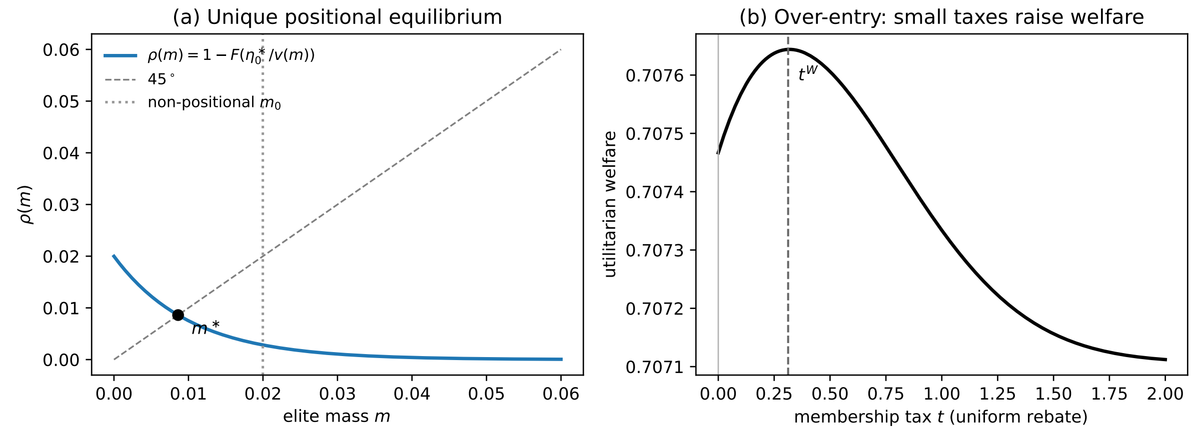

Figure 2: The positional equilibrium. Panel (a): the map ρ of (14) is strictly decreasing, giving a unique elite mass m^*, strictly below the non-positional mass m_0 (dotted). Panel (b): utilitarian welfare as a function of the membership tax t (uniform lump-sum rebate); welfare rises for all small taxes — the signature of over-entry — and peaks at t^W. Parameters as in Section 6 with $v(m) = 1/(1 + 25m)$.

7.5 Numerical continuation

Continuing the example of Section 6 with $v(m) = 1/(1+\kappa m)$, $\kappa = 25$: the elite mass falls from $m_0 \approx 0.0200$ to $m^* \approx 0.0086$ ($v(m^*) \approx 0.82$) (dilution cuts the elite by more than half) and output rises to $\bar{y} \approx 0.847$. The same FOSD taste shift that raises the baseline elite mass by 0.0085 raises the positional mass by only 0.0023. Utilitarian welfare is strictly increasing in the membership tax at $t = 0$; the welfare-maximizing tax in the example is $t^W \approx 0.31$ (about 5.4% of the elite manager's net income, and close to the first-order Pigouvian level $t^* \approx 0.35$), under which the elite mass falls to 0.0050 and output rises further. Figure 2 displays the fixed point and the welfare curve. All computations are verified in `analysis/positional_model.py`.

8 Dynamics: Bequests, Growth, and Rational Impoverishment

The static analysis holds wealth fixed. This section lets the economy accumulate: each generation inherits the resources its parents saved, plays the club economy of Sections 4–7 within its lifetime, and bequeaths. Three questions organize the analysis. Does the market for elite status expand or contract as an economy grows? Does the elite sector's resource burn feed back into accumulation? And, the question any reader of Section 5 will ask, if status spending erodes the wealth of future generations, do rational, forward-looking agents really let it happen?

8.1 The overlapping-generations economy

Time is discrete, $t = 0, 1, 2, \ldots$. Each period a measure-one generation is born, inherits per-capita resources k_t (the input good), and lives for one productive period within which the schooling-then-employment sequence of the static model unfolds (memberships are dated within the period, as in Ellickson et al. 2006). Preferences extend (1) with a warm-glow bequest made in the *produced* good — output invested today is the resource base of the next generation:

$$u \;=\; d(\ell;\eta)\, x_1^{\zeta/2}\, x_2^{\zeta/2}\, b^{\,1-\zeta}, \qquad \zeta \in (0,1), \tag{16}$$

so an agent with net income w spends $(\zeta/2)w$ on each consumption good and $(1-\zeta)w$ on the bequest, and attains indirect utility $d\, w\, \psi(p_2)$, where $\psi(p_2)$ is a price index common to all agents (Lemma 3) — *still linear in net income.* Every occupational-indifference condition, membership price, the output price (5), the cutoff (7), and the positional fixed point (14) therefore carry over verbatim at wealth k_t; only the competent-sector scale n adjusts for the saving margin, and $\zeta = 1$ recovers the static model exactly (Lemma 3 in Appendix D). Invested output converts into next-period resources at a rate $\theta > 0$ (a constant-returns storage activity, freely operated, whose linearity makes it profitless and ownerless; the input good itself perishes at the period's end), and children inherit the economy-wide average bequest:

$$k_{t+1} \;=\; \xi(k_t) \;\equiv\; \theta\,(1-\zeta)\,\frac{\bar{w}(k_t)}{p_2(k_t)}, \tag{17}$$

where $\bar{w}(k)$ is aggregate net income — endowment plus the value added of the schools and firms that form at wealth k. Uniform inheritance abstracts from within-generation heterogeneity in inheritances (the gift is pooled, so there are no dynasties in the model, only generations); the Galor–Zeira-style interaction of inherited inequality with elite membership is an important extension we defer.

Definition 1. Given $k_0 > 0$, a *dynamic equilibrium* is a sequence of period allocations and prices such that, in every period t: the allocation is a group equilibrium of the period economy at wealth k_t — the interior equilibrium of Lemma 1 (or its positional counterpart) when Assumption 1 holds at k_t, and the elite-free equilibrium when $z(k_t) \leq 0$ — and k_{t+1}

is given by (17).

Because preferences (16) contain no forward-looking argument, the definition is genuinely recursive: no agent's optimization requires forecasting, and each period's prices depend only on k_t.

Proposition 6 (The transition map). *Let $K \subset \mathbb{R}_+$ be an interval on which the period equilibrium of Definition 1 is well defined (Assumption 1 where the elite sector is viable, the elite-free interior configuration elsewhere). Under (16): (i) the within-period allocation at wealth $k_t \in K$ is the club equilibrium of Section 5 (or Section 7) at k_t; (ii) ξ is continuous on K, with $\xi'(0^+) = \theta(1-\zeta)\phi/(c+r)$, and the elite-free map is bounded (diminishing returns operate through the price of investment, $p_2(k)$, which rises with wealth); (iii) if $\theta(1-\zeta)\phi > c + r$, the origin repels, and any interior crossing of the diagonal from above is a steady state, stable whenever $|\xi'| < 1$ there.*

In our parameterization K comfortably contains the steady states and transition paths studied below, each crossing is unique with $|\xi'| < 1$, and at substantially higher wealth or taste the interior configuration eventually fails (the population constraint binds), at which point the map would have to be extended by the corner regimes, which we do not need. The recursivity of Definition 1 answers the "do they really let it happen?" question in its sharpest form: not only are expectations correct along the path — they are not even required. The path (k_t) generated by (17) is common knowledge, and every generation can compute the decline it participates in. No individual would behave differently with perfect foresight: each is atomistic, her gift cannot move the aggregate, and her within-period choices are optimal at the prices she faces. The impoverishment we exhibit below is not a mistake, a bubble, or a failure of anticipation; it is what optimizing behavior looks like when status is among the things people want. We are careful about what this does and does not establish: with warm-glow preferences the absence of an expectational channel is a feature of the specification. Appendix E shows that the feature is not load-bearing. The paper's status margin (prices, wages, the cutoff, and the elite mass at every wealth level) is invariant to the transfer motive: any savings rule whatsoever, forward-looking or not, changes only the volume of saving and hence the speed and destination of the wealth path, never the within-period mechanism (Proposition 9). And warm glow is not an evasion but a minimal requirement: under pooled inheritance, purely altruistic parents free-ride completely and transfer nothing (Proposition 10), so some direct gift motive is necessary for the economy to persist at all. The natural sharpening, dynasty-specific transfers with forward-looking parents, is discussed in Section 10.

8.2 Status is a luxury

Proposition 7 (Luxury). *The elite cutoff $\eta_0^*(k) = \mu k/z(k) - 1$ is strictly decreasing in k — and hence the elite mass $m^*(k)$ strictly increasing in k wherever the elite sector is viable — if and only if $\alpha(c+r) < \beta c + \tau r$: precisely the resource-productivity condition* (11) *of Proposition 2.*

The mechanism is instructive. The elite sector's excess costs — the school's burn $(\tau - 1)r$, the firm's burn $(\beta - 1)c$ — are *fixed* resource costs, while every wage in the economy scales with wealth. As k grows, the burn shrinks relative to the status discount that finances it, so the marginal entrant needs less taste for status to justify entry: rich societies buy more eliteness not because their preferences differ but because they can better afford the waste. Below a viability threshold ($z(k) \leq 0$) the elite sector cannot break even at any taste level: in this economy of identical endowments, a society too poor to cover the burn has no market for wasteful status. (With endowment inequality, a wealthy stratum can sustain such a market at low *average* wealth — historical markets in offices, commissions, and titles are the obvious examples — so the proposition is a within-economy comparative static on aggregate wealth holding the distribution fixed.) No nonhomotheticity is assumed in *preferences*; the luxury property is technological, produced by fixed real resource requirements meeting wages that scale with wealth, and it would attenuate if the elite sector's costs scaled with wealth as well — an empirical question about the cost structure of elite institutions. That the operative condition is the same one that makes the elite sector a drag on output is what disciplines the mechanism.

8.3 The status drag and the spread of taste

Proposition 8 (Status drag). *At any wealth k at which the elite sector is active,* sign $\partial\xi/\partial m^* =$ sign$[\alpha(c+r) - (\beta c + \tau r)]$. *Under condition* (11)*, therefore: (i) the transition map lies strictly below the no-elite-sector map wherever the elite is active, so that whenever each map crosses the diagonal once in that region, the steady states are ordered $k^*_{\text{elite}} < k^*_{\text{none}}$; (ii) a FOSD spread of the status-taste distribution, strict at the relevant cutoffs, lowers ξ pointwise (strictly where the elite is active), and under the same single-crossing proviso strictly lowers the steady state; (iii) under positional status, and for taste shifts satisfying the damping hypothesis of Proposition 3(ii), the pointwise decline of ξ is strictly smaller at every wealth level.*

One condition (the elite sector produces less output per unit of resources) now governs all three of the paper's mechanisms: the static output decline (Proposition 2), the luxury property (Proposition 7), and the dynamic drag (Proposition 8). In the example every proviso holds, and the three steady states order as the propositions suggest: $k^*_{\text{abs}} < k^*_{\text{pos}} < k^*_{\text{none}}$. Together they compose the paper's dynamic narrative: growth makes status affordable, status absorbs the resources that fund growth, and the economy converges to a steady state permanently poorer than its status-free counterfactual — with every generation optimizing, and every generation aware.

Two features of the dynamics deserve emphasis. First, there is no collapse: since the elite sector is unviable at low wealth (Proposition 7), the map near the origin is elite-free and the origin repels (Proposition 6). Status spending impoverishes; it does not extinguish. The "self-destruction of the economy" takes the form of a permanently lower steady state, not a death spiral — and the mechanism that limits the damage is the same one that creates it: as wealth falls, the economy can no longer afford its elite. Second, positional dilution is again protective, now dynamically: because entry dilutes status, the elite mass responds far less to spreading taste, and steady-state wealth barely moves. Societies in which status is relative are (by that very relativity) partially immunized against the status drag.

On intertemporal welfare we claim less, deliberately. Within each period, the absolute-status allocation is Pareto optimal given k_t (Theorem 1); but future generations are not parties to any period's core, warm-glow bequests (Andreoni, 1990) internalize the inter-generational margin only partially, and overlapping-generations economies admit dynamic inefficiency of the Diamond (1965) type. We emphasize accordingly that the steady-state comparisons above are statements about *wealth*, not welfare: if the warm-glow path over-accumulates, a lower steady state need not be a worse one. Whether the equilibrium path is efficient among feasible paths — and by what criterion, given that the glow itself is a utility flow — is a question we flag for the sequel rather than resolve here.

8.4 Numerical continuation

Continuing the running example with $\zeta = 0.8$ (a 20% bequest share) and θ calibrated so that the absolute-status steady state is exactly the static example's wealth, $k^* = 2$: the steady state is stable ($\xi'(k^*) = 0.17$), with the static example's prices, wages, cutoff, and elite mass reproduced as the steady-state allocation. The no-elite counterfactual steady state is 2.085; the positional steady state 2.045. As the taste scale rises from $s = 0.10$ to $s = 0.19$, the absolute-status steady state falls to 1.73 — a 16.8% wealth loss relative to the no-elite benchmark, with the elite sector (managers together with their firms' workers) swelling to about a fifth of the population, while the positional steady state falls only to

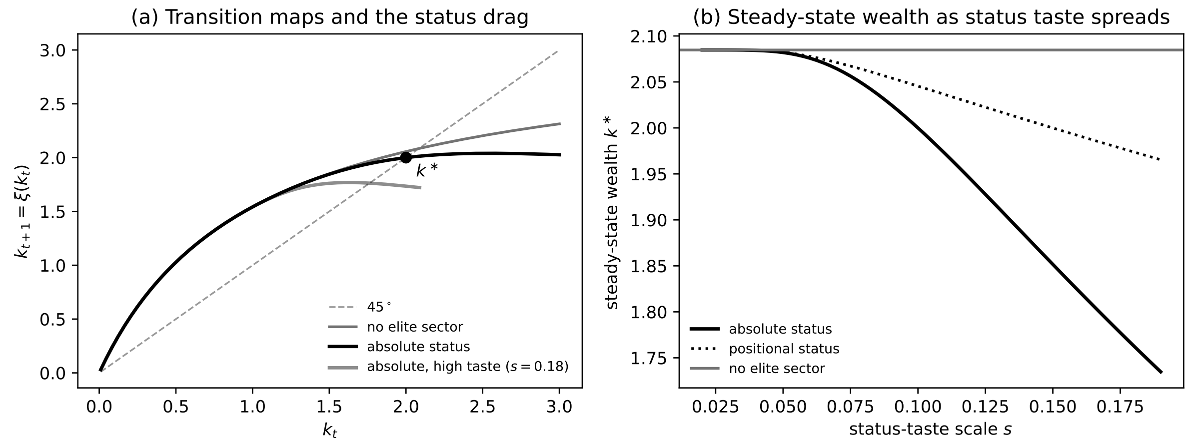

Figure 3: Dynamics. Panel (a): transition maps at baseline taste ($s = 0.1$) for the no-elite counterfactual and the absolute-status economy, and the absolute-status map after a taste spread ($s = 0.18$); the marked point is the calibrated steady state $k^* = 2$. Panel (b): stable steady-state wealth as the taste scale s rises — the absolute-status economy is impoverished by the spread of status taste, the positional economy is largely protected by dilution. Parameters as in Sections 6 and 7 with $\zeta = 0.8$, $\theta = 5.141$.

1.97 (a 5.7% loss). After a taste shift to $s = 0.18$, the transition $2.00 \to 1.74 \to 1.77 \to 1.76$ settles within three generations; the mild oscillation reflects the map's locally negative slope at the new steady state ($\xi' \approx -0.08$): a higher inherited k supports a larger elite, which shrinks the aggregate gift — the luxury mechanism operating period by period. Figure 3 displays the maps and the steady-state decline. All computations are verified in `analysis/dynamic_model.py`.

9 Empirical Content

The model was built to explain a pattern, and it returns a set of predictions sharp enough to be wrong. We collect them here, together with the existing evidence that bears on each.

Compensation. Corollary 1 reproduces the configuration of conspicuous gross salaries alongside a predicted (not yet measured) inferior net position, and its comparative statics are sharper than the accounting identity suggests. Because the elite wage is pinned by the firm's revenue, a change in elite tuition is absorbed *one for one by the status discount*: the equilibrium gross wage satisfies $-q_{4.1} = \alpha p_2 \phi - \beta c + (3 - 2\lambda^e)k - k$, in which τ does not appear (machine-verified). The model therefore predicts *zero incidence of elite tuition on elite salaries* — where a competitive human-capital account, in which wages must recoup education costs on the margin, predicts positive pass-through. Tuition-shock episodes in full-cost credential markets (elite professional degrees, where price approximates resource cost, unlike subsidized undergraduate education) are thus the discriminating experiment. The second prediction is the net one: returns to the elite career track computed net of the full cost of elite education should fall below the non-elite track's — a claim no current estimate supports or refutes, and one the best causal evidence pressures: admission effects large enough to move mean earnings (Chetty et al., 2026) and top-percentile odds (Zimmerman, 2019) exceed any plausible tuition differential, which through the lens of the identity says observed premia carry scarcity rents or bundled ability on top of the status-financed component. The decomposition then bounds the status component; it does not cap the premium.

Firm accounts versus resource accounts. In the central configuration, every firm-level comparison flatters the elite sector: higher output per firm, better-paid managers. The drain appears nowhere on any firm's books, because it is incurred in the school system (Proposition 2); and it need not appear as a fall in measured GDP either, since national accounts value education output at input cost, so the burn is booked as education-sector value added. The model-consistent observable is therefore not aggregate TFP but the *resource cost per elite position*: the quantity of inputs absorbed by credentialing institutions per managerial slot filled, which the model predicts rises with wealth and with the spread of status taste while firm-level performance metrics improve. Credential inflation (more resources devoted to obtaining a given position) is this wedge in labor-market form, and it is measurable where TFP decompositions are not. We note the magnitude honestly: elite credentialing is a small share of aggregate resources in any modern economy, so the mechanism's aggregate bite in our stylized examples (double-digit steady-state losses) reflects the model's deliberately outsized elite sector, not a calibration.

Status as a luxury. Proposition 7 predicts that the elite share of the population rises with aggregate wealth, holding tastes and the endowment distribution fixed. The prediction is a within-economy comparative static, not a cross-country ranking: because our agents share a common endowment, the model's "poor economy" is one in which *no one* can cover the burn, whereas historically poor societies with wealthy strata ran flourishing markets in skill-free position (venal offices, purchased commissions, sold degrees) exactly as an endowment-heterogeneous version would predict. The mechanism the proposition isolates — a fixed resource burn financed by wages that scale with wealth — implies that within an economy, growth expands the elite share, and ties that expansion to the same condition that governs the aggregate drag.

Distinguishing absolute from positional status. The paper's two welfare regimes are observationally similar in a cross-section (wasteful elite institutions persist in both) but differ in their dynamics. Positional status implies self-damped responses of the elite share to demand shifts (Proposition 3) and predicts that expansions of elite institutions dilute the premium attached to them; absolute status implies proportional pass-through. Historical episodes of elite-seat expansion therefore identify the operative regime, and with it the policy conclusion, since only the positional regime admits a welfare-improving membership tax (Proposition 5).

What the causal evidence can and cannot say. The best-identified estimates of elite-education effects compare admitted with rejected applicants and find gains concentrated in top-tail outcomes and access to prestigious positions, large enough to move mean earnings (Zimmerman, 2019; Chetty et al., 2026). Such designs identify *relative* effects and cannot distinguish the creation of output from the reallocation of positions. The model predicts exactly this pattern under pure reallocation: elite membership causally moves its holder up the queue for a fixed set of elite seats while producing nothing — so large private returns to admission coexist with zero (or negative) aggregate returns to elite-sector expansion. The aggregate margin, not the individual one, is where the model's distinctive content lies.

10 Discussion

GDP versus welfare. Propositions 1–2 and Theorem 1 jointly formalize a distinction the public debate conflates: what declines as elite status spreads is what statistical agencies measure, not what agents maximize. Any diagnosis of "inefficiency" from output or TFP data alone presumes status utility does not count — a normative stance that should be argued, not assumed.

Positional scarcity and rents. Positional status supplies a natural amplification channel for observed premia: with v strictly decreasing, the elite characteristic is in endogenously limited supply, and extensions in which firms compete for a capped pool of incumbent elites would shift $q_{4.1}$ further toward scarcity rents (widening observed premia beyond tuition reimbursement) without disturbing the marginal entrant's status discount.

Extensions. Three extensions strike us as most valuable. First, dynasty-specific inheritance: uniform inheritance isolates the aggregate mechanism, but letting parents bequeath directly to their children would interact inheritance with elite membership in the manner of Galor and Zeira (1993), and would let the model speak to intergenerational mobility at the top. The sign is worth stating honestly: because elite managers accept lower *net* incomes, dynasty-specific bequests would transmit less wealth down elite lines — the status discount is dynastically self-impoverishing — so the model's persistence is institutional rather than dynastic, and reconciling the two is precisely what the extension is for. Relatedly, replacing warm glow with forward-looking altruism would make the rationality of decline a theorem about expectations rather than a feature of the specification; under pooled inheritance, altruistic parents face a commons in the aggregate gift, which plausibly strengthens the drag, though we have not proven this. Second, parents who value their children's *status* rather than the bequest itself — the natural sharpening of the warm-glow specification, which would make elite membership itself heritable in demand. Third, intertemporal welfare: each period's absolute-status allocation is optimal given its inheritance, but the path's efficiency across generations — and what, if anything, a generation-spanning planner could improve even in the absolute-status world — is open, and the answer plausibly differs between the absolute and positional regimes.

11 Conclusion

Elite institutions that consume more than they return are not an affront to the theory of competitive markets; they are an application of it. When status is a priced membership, competition sustains the institutions that produce it for the same reason it sustains any other industry: willing buyers cover the costs. This paper has priced elite status explicitly, in the most demanding competitive framework available, and traced the consequences through compensation, output, welfare, and growth. The results discipline both sides of a familiar argument. Against the presumption that visible elite premia and superior elite-firm performance certify productive efficiency, the model shows both arising in an economy that competition is making permanently poorer. Against the presumption that elite waste is a market failure demanding correction, the model shows the waste is Pareto optimal — unless status is valued for its scarcity, in which case, and only in which case, there is over-entry and a corrective tax that raises output and welfare together. The operative empirical question is therefore not whether elite institutions are wasteful, but why their members want them: for what membership is, or for what it is not shared with. Markets work either way; material efficiency follows in neither.

A Derivations and Proofs

All computations in this appendix are verified symbolically in `analysis/static_model.py`.

A.1 Demand and indirect utility

An agent with net income w solves $\max_{x_1,x_2} d\sqrt{x_1 x_2}$ subject to $x_1 + p_2 x_2 = w$, giving $x_1 = w/2$, $x_2 = w/(2p_2)$, and indirect utility $d\,w/(2\sqrt{p_2})$.

A.2 Dominated lists

Lemma 2. *At any prices satisfying (4) with $r, \tau r > 0$, every feasible list other than the five in Section 4 is strictly dominated.*

Proof. The consumption sets allow at most one school and at most one job, and a graduate of one school type cannot hold the other type's managerial membership, so the only feasible lists beyond the five are schooling with no job and schooling combined with a laborer job. (i) Schooling with no job yields taste factor $d \leq \delta_s < 1$ and net income $k - q_j < k$, dominated by leisure. (ii) Schooling plus a laborer job multiplies the laborer's taste factor by $\delta_s < 1$ and subtracts tuition, dominated by the same laborer job alone. □

A.3 Proof of Lemma 1

Budget balance. Schools: $q_1 - r = 0$, $q_2 - \tau r = 0$ by (4). Competent firms: $q_{3.1} + 2q_{3.2} - c + p_2\phi = (k - r - \mu k) + 2(k - \lambda k) - c + p_2\phi = 0$ by (5). Elite firms: $q_{4.1} + 2q_{4.2} - \beta c + p_2\alpha\phi = 0$ reduces, using (5), to $w_E = z$, which is (7). *Optimization.* By Lemma 2 only the five vocational choices need comparison; by construction of (2) and (3), types below η^* are indifferent among the four non-elite choices and strictly prefer them to the elite path, types above strictly prefer the elite path, and the marginal type is indifferent. *Consistency and material balance.* The stated measures fill every group type in the required proportions; market clearing (8) solves

$$n \;=\; \frac{k - (1-3m^*)\frac{k}{2} - m^*\left[\frac{w_E}{2} + w_{L^*} + \tau r + \beta c\right]}{\frac{w_M}{2} + w_L - \frac{3k}{2} + r + c}, \tag{18}$$

positive under Assumption 1. *Walras.* Substituting (18) and (7) into the y_2 market, demand $\sum(\text{measures}) \times w/(2p_2)$ equals supply $\phi(n + \alpha m^*)$ identically (machine-verified).

Uniqueness of the interior configuration follows because (2)–(7) are implied by indifference and budget balance whenever all memberships are active, and (18) is linear in n. $\square$

A.4 Proof of Proposition 1

Subtracting the two firm budget-balance conditions, $p_2\phi(1-\alpha) = (w_M - w_E) - 2(w_{L^*} - w_L) - (\beta-1)c - (\tau-1)r$ after substituting (4); the left side is positive iff $\alpha < 1$, given $p_2 > 0$. $\square$

A.5 Proof of Proposition 2

η^* solves (7), which contains no object depending on F; a FOSD shift weakly raises $1 - F(\eta)$ pointwise, and strictness at η^* raises $m^* = 1 - F(\eta^*)$ strictly. Differentiating $\bar{y} = \phi(n + \alpha m^*)$ with n from (18),

$$\frac{d\bar{y}}{dm^*} = \phi\left(\alpha - 1 + \frac{d(n+m^*)}{dm^*}\right), \qquad \operatorname{sign}\frac{d\bar{y}}{dm^*} = \operatorname{sign}\big[\alpha(c+r) - (\beta c + \tau r)\big],$$

where the second equality is by direct computation (machine-verified in `analysis/static_model.py`): $\frac{d\bar{y}}{dm^*} = \frac{\phi\,[\alpha(c+r)-(\beta c+\tau r)]}{2c+2r+(\mu+2\lambda-3)k}$, whose denominator is positive. $\square$

B Verification of Framework Assumptions

Consumption sets. $X_a = \mathbb{R}^2_+ \times \Lambda_a$ with Λ_a the lists satisfying the schooling prerequisites and the at-most-one-school, at-most-one-job bounds; the list bound is $\bar{\ell} = 2$. Free upward disposal in private goods holds. The correspondence $a \mapsto X_a$ is constant in a (types differ only through u), hence measurable.

Utility. $u(x, \ell; \eta)$ is continuous, strictly increasing in x on the interior of $\mathbb{R}^2_+$ for each ℓ, and jointly measurable since $\eta \mapsto d(\ell;\eta)$ is continuous and $a \mapsto \eta_a$ measurable. One boundary caveat deserves note: $h = \sqrt{x_1 x_2}$ is constant along the axes, so the strict-monotonicity assumption of Ellickson et al. (2006) fails there; their arguments use monotonicity only through local nonsatiation and budget exhaustion, which u satisfies everywhere on $\mathbb{R}^2_+$, so the invocation of their Theorem 1 is unaffected.

Desirable endowments. $u_a(e_a, 0) = h(k, 0) = 0$. If $u_a(x,\ell) > 0$ then $x \gg 0$, so any reduction $x' \le x$, $x' \ne x$, remains in X_a (consumption sets impose no lower bound above zero); hence endowments are desirable in the sense of Ellickson et al. (2006, Section 9.4), and their Theorem 1 delivers membership of every equilibrium state in the strong core and strong Pareto optimality.

Existence. Not invoked: Lemma 1 exhibits equilibrium in closed form. (For non-interior parameter configurations, existence follows from Theorem 2 of Ellickson et al. 2006 upon verifying group irreducibility; this is routine but omitted until needed.)

C Positional Extension: Proofs

All computations are verified in `analysis/positional_model.py`.

C.1 Proof of Proposition 3

The occupational-indifference conditions for non-elite choices do not involve m, so (2) and (5) are unchanged, as is the elite break-even income $w_E = z$ from firm budget balance. The cutoff type solves $(1+\eta v(m))\,\delta_s\delta_m w_E = k$, giving (13); sorting is strict above and below the cutoff because (12) is strictly increasing in η for fixed m. Consistency of m with sorting is (14). Since v is continuous and strictly decreasing and F is continuous and strictly increasing on its support, ρ is continuous and weakly decreasing — strictly wherever $\rho > 0$; when F has bounded support, ρ vanishes identically for m large enough that $\eta_0^*/v(m) \geq \bar{\eta}$. Hence $\rho(m) - m$ is continuous and strictly decreasing with $\rho(0) - 0 = 1 - F(\eta_0^*)$ and $\rho(1) - 1 < 0$, so it has exactly one root $m^* \in [0, 1)$; $m^* > 0$ iff $\rho(0) > 0$, i.e., iff the support of F extends beyond $\eta_0^* = \eta_0^*/v(0)$. (i) Since $v(m^*) < 1$ for $m^* > 0$, $\eta^* = \eta_0^*/v(m^*) > \eta_0^*$, so $m^* = 1 - F(\eta^*) < 1 - F(\eta_0^*) = m_0$. (ii) Replacing F by $\tilde{F} \leq F$ shifts ρ up pointwise, strictly at m^* if the shift is strict at the positional cutoff; with $\rho - \mathrm{id}$ strictly decreasing, the root rises strictly. The baseline mass rises by $\Delta m_0 = F(\eta_0^*) - \tilde{F}(\eta_0^*)$, while the positional mass rises by at most the vertical shift of ρ at the old fixed point, $\Delta\rho(m^*) = F(\eta_0^*/v(m^*)) - \tilde{F}(\eta_0^*/v(m^*))$, because the movement along the strictly decreasing $\rho - \mathrm{id}$ can only shrink the vertical displacement. When the shift satisfies $\Delta\rho(m^*) \leq \Delta m_0$ — the hypothesis of (ii), which holds for the exponential scale family since its vertical shift $e^{-\eta/s} - e^{-\eta/s'}$ is decreasing in η beyond a threshold below the relevant cutoffs — the positional increase is strictly smaller. (iii) Immediate, since given m^* the remaining conditions coincide with those of Lemma 1. $\square$

C.2 Proof of Proposition 4

Work at the equilibrium of Proposition 3. Shutting one elite triple and moving its three members to leisure at their reservation utility changes aggregate excess demand by (a_e, b_e) in (input, output):

$$a_e = \tau r + \beta c + \tfrac{w_E}{2} + w_{L^*} - \tfrac{3k}{2}, \qquad b_e = -\alpha\phi + \frac{w_E + 2w_{L^*} - 3k}{2p_2},$$

and elite-firm budget balance implies $a_e + p_2 b_e = 0$. Opening one competent triple staffed from the (indifferent) leisure pool changes excess demand by $(-a_c, b_c)$ with $a_c = c + r + (w_M + 2w_L - 3k)/2$, $b_c = \phi - (w_M + 2w_L - 3k)/(2p_2)$, and competent budget balance implies $a_c = p_2 b_c$. Both vectors are orthogonal to $(1, p_2)$, hence collinear; moreover $a_e = \frac{1}{2}(\tau r + \beta c + \alpha p_2 \phi) > 0$ and $a_c = \frac{1}{2}(c + r + p_2\phi) > 0$ (both identities machine-verified), so shutting mass $dm > 0$ of elite triples and opening $dn = (a_e/a_c)\, dm > 0$ competent triples restores material balance in both goods *exactly*, and Assumption 1(iv) ($\sigma_0 > 0$) supplies the leisure pool from which the $3\,dn$ additional competent-sector members are drawn at reservation utility. Two groups of agents require care. Non-elite agents, and the laborers released from the shut elite firms, are indifferent across all their activities at equilibrium prices, so their reassignment is utility-neutral. The shut firms' *managers* — types $\eta \in (\eta^*, \eta_{dm})$, where $F(\eta_{dm}) - F(\eta^*) = dm$ — are not: each strictly preferred elite membership, with an individual surplus that vanishes as $\eta \downarrow \eta^*$, so aggregate mover surplus is of order dm^2 while the de-dilution gain to the $m^* - dm$ remaining elites is of order dm. Compensate each mover with a goods transfer equal to her forgone surplus, financed pro rata by the remaining elites: for dm small the total compensation is second order, the elites' first-order gain $\eta\,[v(m^* - dm) - v(m^*)]\,\delta_s \delta_m w_E\, h_0 > 0$ strictly covers it, and every mover is exactly indifferent. Finally, transfer an arbitrarily small additional amount of each good from the (strictly better off) elites to all agents pro rata: by continuity the elites remain strictly better off, and strict monotonicity on the interior makes almost every agent strictly better off. The resulting state is feasible and strictly Pareto dominates the equilibrium state; the grand coalition can implement it, so the state is blocked and not in the core. □

C.3 Planner cutoff condition, the wedge (15), and Proposition 5

Consider the one-parameter family of feasible states indexed by the cutoff $\hat{\eta}$, constructed as in the proof of Proposition 4 (non-elite members held at reservation utility, displaced managers compensated for their marginal surplus, material balance restored via the competent margin). Elite utility is $U(\eta,\hat{\eta}) = (1+\eta v(m(\hat{\eta})))\,\delta_s\delta_m w_E\,h_0$ with $m(\hat{\eta}) = 1-F(\hat{\eta})$ and $h_0 \equiv 1/(2\sqrt{p_2})$. Differentiating aggregate elite utility and using $m'(\hat{\eta}) = -F'(\hat{\eta})$:

$$\frac{d}{d\hat{\eta}}\int_{\hat{\eta}}^{\infty} U\,dF = -\underbrace{U(\hat{\eta},\hat{\eta})F'(\hat{\eta})}_{\text{marginal type's utility}} \underbrace{+F'(\hat{\eta})\,|v'(m)|\,\delta_s\delta_m w_E h_0 \int_{\hat{\eta}}^{\infty}\eta\,dF}_{\text{de-dilution of inframarginals}}.$$

At an interior optimum the marginal type's *surplus* over reservation utility equals the second term divided by $F'(\hat{\eta})$; converting to income units by dividing by the marginal type's marginal utility of income $(1+\hat{\eta}v)\delta_s\delta_m h_0$ gives the wedge (15). At laissez-faire the marginal surplus is zero while the wedge is strictly positive: over-entry. For Proposition 5: (i) with tax t, the rebate feeds back into all incomes, so the equilibrium is a joint fixed point in the elite mass and the rebate; at fixed rebate the map remains continuous and strictly decreasing, so uniqueness persists at that level, and in the example the joint fixed point is unique with $m(t)$ strictly decreasing (the companion script solves the joint problem explicitly). (ii) The over-entry statement is the planner cutoff condition derived above. We emphasize what is *not* claimed: the derivative of utilitarian welfare at $t=0$ does not reduce to the de-dilution wedge, because the rebate shifts every agent's effective income (moving all wages and prices at first order) and redistributes between agents with unequal marginal utilities of income. In the example, $dW/dt|_{t=0^+} = 0.00125$, of which the de-dilution term accounts for 0.00058; the remainder is the rebate's general-equilibrium and redistributive contribution. The wedge (15) characterizes over-entry along the planner's cutoff family; it is not the welfare-maximizing tax, which in the example is $t^W \approx 0.31$ against a wedge of ≈ 0.35. (iii) In the example, $dm/dt < 0$, and with condition (11) this gives $d\bar{y}/dt > 0$ by Proposition 2. The global welfare-maximizing tax in the numerical example, which accounts for all equilibrium feedback including the rebate, is computed in `analysis/positional_model.py`. □

D Dynamics: Proofs

All computations are verified in `analysis/dynamic_model.py`.

Lemma 3 (Period equivalence). *Under preferences (16), an agent with net income w facing p_2 chooses $x_1 = (\zeta/2)w$, $p_2x_2 = (\zeta/2)w$, $p_2b = (1-\zeta)w$, and attains indirect utility $d\,w\,\psi(p_2)$, where $\psi(p_2) \equiv (\zeta/2)^{\zeta}(1-\zeta)^{1-\zeta}p_2^{-(1-\zeta/2)}$ is a price index common to all agents. Hence all occupational-indifference conditions, membership prices (4), the output price*

(5), the cutoff (7), and the positional fixed point (14) hold at wealth k_t exactly as in the static sections. Market clearing in the input good reads $(\zeta/2)W+n(r+c)+m^*(\tau r+\beta c)=k$, which is linear in n; output clearing holds identically given it (Walras, machine-verified), and at $\zeta=1$ the static condition (8) is recovered. $\square$

D.1 Proof of Proposition 6

(i) is Lemma 3. (ii): aggregate income satisfies $\bar{w}=k+n(\mu+2\lambda-3)k+m^*\varepsilon$ with $\varepsilon\equiv w_E+(2\lambda^e-3)k$; substituting the clearing solution for n gives $\bar{w}(k)$ in closed form, and $\xi=\theta(1-\zeta)\bar{w}/p_2$ with $p_2(k)\phi=c+r+(\mu+2\lambda-3)k$ from (5). As $k\to 0^+$ the elite sector is inactive (Proposition 7) and $\bar{w}\to k$, so $\xi/k\to\theta(1-\zeta)\phi/(c+r)$; as $k\to\infty$, n and $\bar{w}/(p_2\phi)$ converge, so ξ is bounded (limits machine-verified). (iii): with $\xi(0)=0$, $\xi'(0^+)>1$, ξ continuous and bounded, ξ crosses the diagonal from above at least once; at any such crossing the steady state is stable if $|\xi'|<1$, verified in the example ($\xi'=0.17$). The no-forecasting claim is immediate: (16) depends on own consumption and own gift only, and period-t prices depend only on k_t. $\square$

D.2 Proof of Proposition 7

$\eta_0^*(k)=\mu k/z(k)-1$ with $z=\alpha p_2\phi-\beta c-\tau r+(3-2\lambda^e)k$. Then $d\eta_0^*/dk=\mu\,(z-kz')/z^2$ and direct computation (machine-verified) gives $z-kz'=\alpha(c+r)-\beta c-\tau r$. Hence η_0^* is strictly decreasing iff $\alpha(c+r)<\beta c+\tau r$, and $m^*=1-F(\eta_0^*)$ (or its positional analogue, whose fixed-point map shifts up as η_0^* falls) is strictly increasing wherever $0<z<\mu k$. Viability: $z(k)\leq 0$ for k below the root of z, where no cutoff exists and the elite sector is empty. $\square$

D.3 Proof of Proposition 8

Note $\varepsilon=w_E+(2\lambda^e-3)k=\alpha p_2\phi-\beta c-\tau r$: the elite triple's value added. Substituting the clearing solution for n into $\bar{w}$ and differentiating at fixed k,

$$\operatorname{sign}\frac{\partial\bar{w}}{\partial m^*}=\operatorname{sign}\Big[(r+c)\,\varepsilon-(\tau r+\beta c)\,(p_2\phi-r-c)\Big]=\operatorname{sign}\Big[p_2\phi\,\big(\alpha(c+r)-\beta c-\tau r\big)\Big],$$

(machine-verified), and $\partial\xi/\partial m^*$ has the same sign since p_2 is fixed given k. Under (11) the sign is negative: (i) follows pointwise and hence for the stable steady state (ξ falls, the crossing with the diagonal moves left); (ii) a FOSD shift raises $m^*(k)$ at every k with the elite active (Propositions 2 and 3(ii)), lowering ξ pointwise; (iii) under positional status the same shift raises m^* by strictly less (Proposition 3(ii)), so ξ falls by strictly less at every k, and the steady states order as stated (verified in the example). $\square$

E The Transfer Motive: Invariance and Necessity

This appendix records two results that locate exactly how much of the dynamic analysis depends on the warm-glow specification (16).

Proposition 9 (Savings-motive invariance)**.** *Fix $k > 0$ and consider any period allocation in which each agent divides net income w between within-period consumption in the fixed proportions of a homothetic aggregator and a transfer of the produced good, with the transfer share given by an arbitrary measurable rule $\varsigma : \mathbb{R}_+ \to (0,1)$ — a function of k, of forecasts of any future variables, or of anything else common to agents. Then the equilibrium membership prices, wages, the output price p_2, the cutoff η^*, and the elite mass are those of Sections 5 and 7, independent of ς; only the competent-sector scale n and the transition $k_{t+1} = \theta\,\varsigma(\cdot)\,\bar{w}(k_t)/p_2(k_t)$ depend on it.*

Proof. Indirect utility remains $d\,w\,\psi_\varsigma(p_2)$ for a price index ψ_ς common to all agents, because the expenditure shares are common; every occupational-indifference condition therefore compares $d \cdot w$ across choices exactly as in Lemma 1, pinning the same wages and the same cutoff, and the firm and school budget-balance conditions are untouched. The transfer share enters only the market-clearing condition, which determines n, and the law of motion. (The independence of the equilibrium wages, p_2, and z from the share parameter is verified symbolically in the companion code.) $\square$

The proposition says that expectations, however sophisticated, have no channel into the paper's mechanism: a fully rational, forward-looking population could differ from our warm-glow population only in how much it saves, and hence in how fast and how far wealth moves, never in how the market for status operates at any given wealth. The decline results are in this sense robust to the transfer motive: any motive generating a wealth path through the region where the elite sector is viable encounters the same luxury property and the same drag.

Proposition 10 (The bequest commons)**.** *Suppose inheritance is pooled as in Section 8 but parents are purely altruistic, valuing their child's equilibrium utility rather than the gift itself. Then the unique equilibrium transfer is zero.*

Proof. A child's endowment is the economy-wide average transfer, so an atomistic parent's own gift has measure-zero effect on her child's utility while reducing her own consumption; the strictly dominant choice is a zero gift, for every parent, at any conjectured aggregate. $\square$

Under pooling, that is, pure altruism cannot sustain the economy at all: some direct motive over the gift (warm glow being the simplest) is necessary, not ad hoc. The configuration in which altruism has traction is dynasty-specific inheritance, which requires heterogeneous wealth and is the extension discussed in Section 10. We note for completeness the remaining variant, life-cycle saving for old-age consumption: it requires the old cohort to own the period's resource endowment, which removes the young's leisure margin and with it the anchor of the wage system, so it is a different economy rather than a robustness check on this one.

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